Ericsson PE Node BTM-PE-05 Hardware/Environmental Fault Resolution Procedure (ALARM_7301)

Node: BTM-PE-05 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_7301

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults on Ericsson PE node BTM-PE-05, specifically addressing ASIC memory parity errors, line card crashes, and power supply redundancy loss. The PE node serves as a critical component in the BTM Layout Aggregation site, responsible for aggregating and forwarding traffic between multiple network segments. The node's hardware and environmental conditions, including temperature fluctuations, power supply redundancy, and ASIC chip quality, can significantly impact its performance and reliability. When an ALARM_7301 is triggered, indicating a hardware/environmental fault, it is essential to perform a thorough investigation to identify the root cause and implement corrective actions to prevent further disruptions to the network.

Pre-requisite Checks

- Verify that the BTM Layout Aggregation site is accessible and safe for dispatching remote hands.

- Check the redundant link status to ensure that it is operational and not contributing to the failure.

- Validate that the Ericsson PE node BTM-PE-05 is powered on and reachable via the management interface.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware defects or manufacturing issues	High
Line card crashes due to software bugs, configuration errors, or resource constraints	Medium
Power supply redundancy loss due to equipment failure or power disruptions	Low

Diagnostic Commands & Expected Outputs

```
Command: show node BTM-PE-05 statistics, Expected Output: node BTM-PE-05 statistics, Description: This command displays the node's statistical information, including error counts, which can help identify potential issues.

Command: show linecard BTM-PE-05 status, Expected Output: linecard BTM-PE-05 status, Description: This command provides the line card's status, including its operational state and any error messages.
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```
Command: show power BTM-PE-05 status, Expected Output: power BTM-PE-05 status, Description: This command displays the power supply's status, including its operational state and any error messages.
```

Resolution Steps

Step 1: Step: Power cycle the Ericsson PE node BTM-PE-05, Command: reboot node BTM-PE-05 force, Description: This step involves forcing the node to restart, which can help resolve issues related to software bugs or configuration errors.

Step 2: Step: Replace the faulty ASIC or line card, Command: replace node BTM-PE-05 asic 1, Description: This step involves replacing the faulty ASIC or line card with a new one, which can help resolve issues related to hardware defects or manufacturing issues.

Step 3: Step: Verify and restore power supply redundancy, Command: verify power BTM-PE-05 status, Description: This step involves verifying that the power supply redundancy is restored and functioning correctly.

Post-Check Verification

```
Command: show node BTM-PE-05 statistics, Expected Output: node BTM-PE-05 statistics, Description: This command is used to verify that the node's statistical information is normal and error-free.

Command: show linecard BTM-PE-05 status, Expected Output: linecard BTM-PE-05 status, Description: This command is used to verify that the line card's status is normal and error-free.

Command: show power BTM-PE-05 status, Expected Output: power BTM-PE-05 status, Description: This command is used to verify that the power supply's status is normal and error-free.
```

Roll-back Procedures

- Step: Restore the previous configuration, Command: restore config BTM-PE-05, Description: This step involves restoring the previous configuration to its original state, which can help prevent further disruptions to the network.

- Step: Revert the faulty ASIC or line card, Command: revert node BTM-PE-05 asic 1, Description: This step involves reverting the faulty ASIC or line card to its original state, which can help prevent further disruptions to the network.